

Water, Water Everywhere but not a Drop to Drink!

The most important resource of life is water. Unfortunately, this is true for the semiconductor business as well. Ultra-pure water is the most crucial component required for manufacture of semiconductor components and chips. The city of Austin, Texas is home to three semiconductor giants: AMD, Motorola and Samsung Austin Semiconductor. In the year 1999, Motorola used 1.78 billion gallons of water, AMD 1.18 billion gallons and Samsung used 353 million gallons. This is just a small example of just how thirsty microchips are. According to figures of Texas Instruments, the cost of designing a single fabrication in 1993 was 591 million gallons of water.

eventually be ingested by humans and animals. It is responsible for colic in children, and long-term effects of lead poisoning also include osteoporosis.

In certain parts of the world, CRT recycling centres have been set up that are catering to localised needs—including the US, UK and Singapore. The CRT recycling process mainly works on separating the heavy metals out of the other monitor wastes that can be disposed safely or recycled. Some systems process one tube every 3 to 5 seconds. The Andela CRT Recycling system is one such system. The main problem here is the dearth of CRT recycling centres in India. An ideal implementation is to make the manufacturer take responsibility of recycling.

Computer dumping

Computer dumping is a place—a visit to a landfill

bished with a few minor repairs, such as wiping of disks and alignment of disk drives.

In India, dumping is yet to be seen on a large scale. This is partly due to the large market of used computer systems. Also, computer products are used for a significantly longer time in such a cost-conscious market as India.

A disturbing aspect of computer disposal is the dumping of PCs in third world or developing countries. The trade of toxic goods is banned by the international Basel Treaty. But the Basel Treaty does not focus on computing specifically and loopholes in the treaty can be exploited. Further, this treaty has not been ratified by the US. Thus, it provides an easy path to third-world countries, which recycle (read: reuse), the components, China and Pakistan are the porters of these components.

“The fruits of our high-tech revolution are improperly disposed of at the end of their life cycle.”

“The products are not designed to be recycled.”

such as lead-free computing, resource conservation and waterless processing. But to a large extent, the industry has done nothing about this resource hogging. The drive to increase performance has made people overlook this issue. The only solution is the development of better production facilities and standards.

Display pollution

Cathode Ray Tubes (CRTs) are the heart and soul of most display devices. While CRT screens provide reliable and safe performance throughout their operating life—as long as their outer structure is not damaged—the inner components of the CRT are quite hazardous. The most notorious component is lead. Monochrome monitors have 2 to 3 per cent of lead, whereas colour monitors have up to 28 per cent lead. The Department of Toxic Substances Control (DTSC), California has designated CRTs as a Universal Waste. It requires special procedures for proper disposal and recycling. The lead contained in the CRT does not harm the user during its lifetime as it is encased in glass. A small CRT may have up to 1.9 Kg to 3.8 Kg of lead. Larger CRTs have proportionally more amounts of lead. Lead leaking from improperly disposed monitors can find its way into ground-water, and may

depressing picture of what many refer to as a computer graveyard. The term used to define these wastes is e-waste. E-waste is a source for more than a thousand hazardous substances such as lead, mercury and beryllium. The rising amount of e-waste is also due to the fact that it is more cost-effective to replace damaged components than to repair them due to the rapid evolution of technology. Reuse is one of the most effective and often the most neglected facets of e-waste. Certain computer components can be refurbished.

Why do people import this scrap? The higher profit margin is the main reason. Unlike centres situated in first world countries, the recycling process does not have a high investment cost for the machines used. Trace elements of gold, silver and platinum are found in a typical computer. These precious metals are recovered and then sold at a high price in the market. The fact that is overlooked by most people is the environmentally lethal procedures that must be carried out in order to recover

The Devils Alchemy: What goes into your computer

The materials inside the ubiquitous white box have a concentration of materials, which are the sure shot recipe for bereavement.

Component	Health Hazard	Environmental Hazard
CRT	Silicosis, Inhalation of hazardous materials	Leaching of heavy metals such as lead or barium into groundwater. Possible release of toxic phosphor
PCB	Exposure to toxic substances such as tin and lead	Emission of toxic substances
Plastics	Exposure to heavy metals	Emission of heavy metals
Connectors	Exposure to Polycyclic Aromatic Hydrocarbons (PAH)	Emission of PAH hydrocarbons
Cartridges	Carcinogen exposure	Toner materials with varied toxicity

Source: Exporting Harm, SVTC.